



Bahir Dar University
Bahir Dar Institute of Technology
Faculty of Computing
Network Design Document

for

**Industrial project on Design and Implementation of a secure Tibebe Gihon
Campus Network Infrastructure**

Submitted to the faculty of computing in partial fulfillment of the requirements for the degree
of Bachelor of Science in Information Technology

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Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

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Signature

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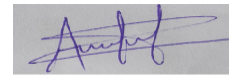
Faculty: Computing

Program: Computer Science

Project Title: Design and Implementation of a secure Tibebe Gihon Campus Network Infrastructure Management System

This is to certify that I have read this project and that in my supervision and the students' Performance, it is fully adequate, in scope and quality, as a project for the degree of Bachelor of Science.

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Name of Advisor

Signature

**Examining committee
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The Role of Group Members

List of Task	List members		
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List of Acronyms

AAA – Authentication, Authorization, and Accounting

ACL – Access Control List

AP – Access Point

ASA – Adaptive Security Appliance

ASW – Access Switch

BGU – Bahir Dar University

BIT – Bahir Dar Institute of Technology

BOQ – Bill of Quantity

CIDR – Classless Inter-Domain Routing

CLI – Command Line Interface

CMHS – College of Medicine and Health Sciences

CPU – Central Processing Unit

DC – Data Center

DHCP – Dynamic Host Configuration Protocol

EMR – Electronic Medical Record

HA – High Availability

ICT – Information and Communication Technology

IOT – Internet of Things

IP – Internet Protocol

IPv4– Internet Protocol Version 4

ISP – Internet Service Provider

L2 – Layer 2 (Data Link Layer)

L3 – Layer 3 (Network Layer)

LAN – Local Area Network

MAC – Media Access Control

MainDC – Main Data Center

MiniDC – Mini Data Center

NAT – Network Address Translation

NOC – Network Operations Center

NS-3 – Network Simulator 3

OS – Operating System

OSPF – Open Shortest Path First

PoE – Power over Ethernet

PT – Packet Tracer

QoS – Quality of Service

RAM – Random Access Memory

RSTP – Rapid Spanning Tree Protocol

SFP – Small Form-factor Pluggable

SIEM – Security Information and Event Management

SNMP – Simple Network Management Protocol

SSID – Service Set Identifier

STP – Spanning Tree Protocol

TGSH – Tibebe Ghion Specialized Hospital

UPS – Uninterruptible Power Supply

VLAN – Virtual Local Area Network

VTP – VLAN Trunking Protocol

WAN – Wide Area Network

WLAN – Wireless Local Area Network

VPN – Virtual Private Network

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Executive Summary

This project presents the design and implementation of a secure, scalable, and reliable campus-wide network infrastructure for Tibebe Ghion Specialized Hospital (TGSH), a tertiary-level teaching and referral hospital affiliated with Bahir Dar University, College of Medicine and Health Sciences. As a major healthcare and academic institution, TGSH supports critical clinical services, medical education, research activities, and administrative operations that rely heavily on uninterrupted and secure network connectivity. The existing network infrastructure at TGSH suffers from limitations related to security, scalability, fault tolerance, and centralized management, creating operational risks and reducing service efficiency. This project addresses these challenges through a structured and modern network design approach.

The primary objective of the project is to design a hierarchical campus network architecture that ensures high availability, strong security, efficient communication, and future readiness. The proposed solution follows a three-layer network model consisting of core, distribution, and access layers. This architecture improves fault isolation, simplifies network management, and enables seamless scalability as the hospital expands its services and user base. An extended star topology is adopted, with the main data center acting as the central hub connecting multiple buildings such as hospital blocks, staff buildings, research centers, library facilities, and student dormitories through high-speed fiber optic backbone links.

To enhance security and performance, the network is logically segmented using Virtual Local Area Networks (VLANs). Separate VLANs are allocated for clinical staff, academic and administrative users, researchers, students, dormitory users, guests, IoT and VoIP systems, servers, and network management. This segmentation reduces broadcast traffic, improves performance, and ensures that sensitive clinical systems are isolated from non-critical user networks. Inter-VLAN routing is centrally implemented at the core layer with controlled access policies to allow only authorized communication between network segments.

Network security is treated as a core design requirement due to the sensitive nature of healthcare and academic data handled by TGSH. A dedicated perimeter firewall is deployed between the internal campus network and the Internet Service Provider to enforce access control policies,

implement Network Address Translation (NAT), and protect the network from external threats. Internal security is strengthened through VLAN-based isolation, restricted management access, and centralized monitoring mechanisms. Guest and dormitory wireless networks are strictly separated from critical hospital systems to prevent unauthorized access and reduce security risks.

Wireless connectivity is integrated into the network design to support mobility for staff, students, and visitors. Enterprise-grade wireless access points powered through Power over Ethernet (PoE) are strategically deployed in high-demand areas such as the library and dormitories. Due to physical distance limitations from the main data center, wireless distribution for dormitory areas is sourced from the library distribution switch, ensuring reliable performance while maintaining centralized security control.

The project also emphasizes effective network management and monitoring. Centralized network management is implemented using standard protocols such as SNMP, SSH, and Syslog to enable proactive fault detection, performance monitoring, and secure administration. A dedicated management VLAN ensures that monitoring and configuration traffic is isolated from user data traffic, improving both security and reliability. Network performance and functionality are evaluated through simulation using Cisco Packet Tracer, allowing validation of connectivity, VLAN segmentation, routing, and security policies before deployment.

Overall, the proposed network design provides TGSU with a robust, secure, and scalable campus network infrastructure that supports its dual role as a healthcare provider and academic institution. The solution improves service continuity, protects sensitive data, enhances operational efficiency, and prepares the hospital for future technological advancements such as cloud-based healthcare services, VoIP communication, and IoT-enabled medical systems. This project serves not only as a practical solution for TGSU but also as a reference model for similar healthcare and academic institutions seeking to modernize their network infrastructure.

Chapter One: Introduction

1.1 Organization Overview

Tibebe Ghion Specialized Hospital (TGSH) is a tertiary-level teaching and referral hospital located in Bahir Dar City, the capital of the Amhara National Regional State, Ethiopia. The hospital was officially inaugurated in November 2018G.C as part of the national initiative to expand access to specialized healthcare services and to strengthen medical education and research in the country.

TGSH operates under the administration of Bahir Dar University, College of Medicine and Health Sciences (CMHS), and serves both as a healthcare service provider and an academic institution.

TGSH was established to deliver advanced clinical services while supporting undergraduate and postgraduate medical education, residency training, and health-related research. The hospital provides a wide range of specialized medical services, including internal medicine, surgery, pediatrics, obstetrics and gynecology, emergency and critical care, radiology, laboratory services, and pharmacy.

In addition to clinical departments, the hospital hosts academic units, research facilities, libraries, staff offices, student dormitories, and administrative departments, all of which depend on reliable information and communication technology infrastructure.

As a large and complex organization, TGSH relies heavily on effective communication and information sharing among its departments and units. Daily hospital operations generate and consume large volumes of data related to patient care, teaching activities, research work, and administrative processes. These operations require continuous access to network services such as internal communication systems, internet connectivity, digital academic resources, and centralized management platforms.

The hospital's mission emphasizes the delivery of high-quality specialized healthcare services, the advancement of medical education, and the promotion of research and innovation. To support this mission, a secure, reliable, and scalable campus-wide network infrastructure is essential. Such a

network serves as the backbone for clinical coordination, academic collaboration, administrative efficiency, and future digital healthcare initiatives.

Given TGSH's dual role as both a healthcare institution and a teaching hospital, the network infrastructure must accommodate diverse user groups, including clinical staff, academic staff, students, researchers, administrators, and guest users. Each group has different access needs, performance requirements, and security considerations. Therefore, the design of a structured and well-managed network is critical to ensuring smooth hospital operations, data protection, service continuity, and readiness for future technological expansion.

History and main activities or business processes running at the organization.

Tibebe Ghion Specialized Hospital (TGSH) is a comprehensive tertiary care and teaching hospital that serves as one of the largest health facilities in Ethiopia. It was affiliated with Bahir Dar University, particularly its College of Medicine and Health Sciences (CMHS).

The hospital was officially inaugurated on November 10, 2018, in Bahir Dar city, the capital of the Amhara National Regional State. The inauguration ceremony was attended by senior federal and regional government officials, demonstrating the national significance of the project.

The establishment of TGSH was part of Ethiopia's strategic effort to expand access to specialized health services and improve medical education infrastructure.

Its construction and development were financed using public resources and infrastructure investment, reflecting its importance as a public property and national health asset. Built on a site of approximately 5–7 million square meters, the hospital was envisioned as a flagship tertiary referral center that would provide specialized medical care, advance research, and strengthen clinical training for future health professionals.

Like most healthcare institutions.

TGSH provides its services in the form of clinical and non-clinical functions. These services are broadly categorized as follows:

1. Emergency and Critical Care Services: The hospital operates a fully functional emergency department that provides 24-hour care for urgent and life-threatening conditions. It is regarded as a regional referral center for emergency and critical care, capable of handling high-acuity trauma cases, medical emergencies, and resuscitation.

2. Inpatient Care With over 500 beds distributed across different wards, TGSB provides admission and treatment for patients in medicine, surgery, pediatrics, obstetrics, and gynecology. Each ward is equipped for specialized services, supported by nursing care and multidisciplinary teams.

3. Surgical and Operating Theatres: The hospital maintains multiple operating rooms equipped to perform both elective and emergency surgical procedures. These theatres support general surgery, orthopedic surgery, neurosurgery, obstetrics, and other specialties. All surgical procedures follow internationally recognized patient safety protocols.

4. Intensive Care Units (ICU/HDU): TGSB has intensive care units and high-dependency units dedicated to the management of critically ill patients. These units are equipped with ventilators, advanced monitoring systems, and specialized staff trained in critical care medicine.

5. Outpatient Clinics Specialty outpatient clinics operate regularly: offering consultations and follow-up services across multiple disciplines. This includes internal medicine subspecialties, surgical clinics, pediatrics, and maternal health.

6. Advanced Diagnostic Services: The hospital provides diagnostic services through its imaging department and laboratory. Imaging includes modern MRI, CT, ultrasound, and X-ray facilities, while the laboratory supports a wide range of clinical testing. These facilities are essential for accurate diagnosis and treatment planning.

7. Teaching, Training, and Research As an academic hospital: TGSB provides clinical training to medical students, residents, and allied health trainees from Bahir Dar University. The hospital

is also a site for clinical research, academic publications, and collaborative projects with national and international institutions.

1.2 Statements of the problem

Tibebe Ghion Specialized Hospital (TGSH) depends on its network infrastructure to support critical clinical services, academic activities, research operations, and administrative functions. As a tertiary care and teaching hospital, the organization requires a reliable, secure, and scalable campus-wide network to ensure uninterrupted communication, protect sensitive information, and support current and future digital healthcare services. However, the existing network infrastructure at TGSH does not adequately meet these requirements.

The current network design lacks a structured and hierarchical architecture. It relies on limited segmentation and centralized components that create single points of failure. Core network devices and backbone links are not sufficiently redundant, which increases the risk of network-wide outages when hardware failures, maintenance activities, or power disruptions occur. Such outages can affect multiple departments simultaneously, disrupting hospital operations and reducing service availability.

From a security perspective, the existing network primarily focuses on perimeter protection, with minimal internal segmentation and weak access control mechanisms. Different user groups—including clinical staff, academic users, students, dormitory residents, and guest users—share overlapping network resources without strict isolation. This increases the risk of unauthorized access, internal misuse, malware propagation, and data exposure. In a hospital environment where confidentiality, integrity, and availability of information are essential, these weaknesses pose serious operational and security concerns.

The current network also suffers from poor scalability and limited support for future expansion. It does not follow a modular design that allows new departments, devices, or services to be added easily.

As a result, integrating emerging technologies such as IoT-based medical devices, centralized hospital information systems, cloud-based services, and advanced network monitoring tools becomes complex and error-prone.

In addition, the absence of centralized network monitoring and management mechanisms limits the ability of ICT staff to proactively detect faults, analyze traffic behavior, and enforce consistent security policies. Network issues are often identified only after users experience service disruptions, leading to delayed response times and prolonged downtime.

These limitations collectively reduce network reliability, increase security risks, and hinder the hospital's ability to deliver efficient healthcare services and academic support. Therefore, there is a critical need for a redesigned campus-wide network infrastructure that is secure, reliable, scalable, and centrally managed. Addressing these challenges through a well-structured network design will enable TGSH to improve service continuity, strengthen internal security, support institutional growth, and prepare for future digital healthcare initiatives.

1.3 Objectives of the Project

This project aims to address the security, reliability, scalability, and performance limitations identified in the existing network infrastructure of Tibebe Ghion Specialized Hospital (TGSH). As a tertiary care and teaching hospital, TGSH requires a robust and secure campus-wide network that supports critical medical operations, academic activities, research, and administrative services while protecting sensitive patient and institutional data.

Accordingly, this project focuses on the design and implementation of a secure, scalable, and reliable network infrastructure that can support current operational demands and accommodate future growth within the scope of the final year project.

1.3.1 General Objective

The general objective of this project is to design a secure, reliable, and scalable campus-wide network infrastructure for Tibebe Ghion Specialized Hospital (TGSH) that ensures efficient communication among departments, protects sensitive clinical and academic data, mitigates internal and external security risks, and supports the hospital's dual role as a healthcare provider and academic institution.

The achievement of this objective will be validated through logical network segmentation, firewall-based security enforcement, centralized monitoring, and simulation-based performance testing.

1.3.2 Specific Objectives

To achieve the general objective, the project focuses on the following specific objectives:

1. Analyze the existing TGSH network infrastructure to identify weaknesses related to security, reliability, redundancy, scalability, and performance.
2. Design a hierarchical and modular campus network architecture based on Core, Distribution, and Access layers to improve fault isolation, manageability, and scalability.
3. Propose a structured Main Data Center and Mini Data Center architecture to support efficient traffic aggregation and localized fault isolation.
4. Implement VLAN-based logical segmentation to separate hospital, staff, library, research, dormitory, guest, and management networks in order to reduce congestion and enhance security.
5. Configure firewall security policies to enforce access control, implement Network Address Translation (NAT), and protect the network from external threats.

6. Control inter-VLAN communication to ensure proper isolation between critical hospital systems and non-critical user networks.
7. Design a secure wireless network by isolating guest and dormitory wireless access from internal hospital systems and deploying PoE-enabled wireless access points in appropriate locations.
8. Establish centralized network monitoring and management using protocols such as SNMP, SSH, and Syslog to enable proactive fault detection and performance monitoring.
9. Evaluate network performance and reliability through simulation and testing using Cisco Packet Tracer, focusing on throughput, latency, and availability under different traffic conditions.
10. Reduce single points of failure within critical network segments to improve service continuity and availability.
11. Design a scalable IP addressing and subnetting scheme that supports current devices and allows seamless integration of future departments, buildings, and services.
12. Ensure future readiness of the network infrastructure by designing it to support emerging technologies such as VoIP communication systems, cloud-based healthcare services, and IoT-enabled medical devices without requiring major architectural changes.
13. Document the complete network design and implementation process in a structured academic format, including requirements analysis, topology diagrams, configuration commands, security policies, and testing results.

1.4 Significance of the project

This project is significant in addressing the growing need for a secure, scalable, and reliable campus network infrastructure at Tibebe Ghion Specialized Hospital (TGSH). As a tertiary healthcare and teaching institution, TGSH depends heavily on effective network communication to support patient care, medical education, research activities, and administrative operations.

From a healthcare perspective, the proposed network design enhances secure communication among clinical departments, helping to protect sensitive patient data and ensure uninterrupted access to critical medical systems. Improved network reliability and fault isolation contribute directly to service continuity, which is essential in a hospital environment where network downtime can impact patient safety.

From an academic and research perspective, the project supports teaching, learning, and research activities by providing stable and high-performance network connectivity for students, instructors, and researchers. The inclusion of structured wireless access and logical network segmentation improves accessibility while maintaining security.

From a technical and professional perspective, the project applies real-world networking concepts such as hierarchical network design, VLAN segmentation, firewall-based security, centralized monitoring, and data center architecture. This strengthens the practical skills and professional competence gained during the internship and final year study.

Additionally, the project provides a reference network design model that can be adapted or extended for similar hospital or campus environments in the future. The documented design and simulation results may assist network administrators, students, and researchers in understanding best practices for designing modern hospital networks significant for the hospital's sustainability and digital transformation efforts.

Finally, the project has broader relevance beyond TGSH. It serves as a practical case study demonstrating how network design principles can be applied in a large public hospital and academic environment. The design approach and documentation can be used as a reference for similar healthcare and educational institutions facing challenges related to network reliability,

security, and scalability. In this way, the project contributes both practically and academically to the field of network design and infrastructure planning.

1.5 Scope of the Project

This final year project focuses on the design and implementation of a secure, scalable, and reliable campus network infrastructure for Tibebe Ghion Specialized Hospital (TGSH). The scope of the study covers the analysis, design, and simulation of a campus-wide network that supports clinical services, academic activities, research operations, and administrative functions within the hospital environment.

The project scope includes the design of a hierarchical network architecture consisting of Core, Distribution, and Access layers, as well as the definition of Main Data Center and Mini Data Center roles within the campus network. The study also covers logical network design aspects such as VLAN-based segmentation, IP addressing and subnetting, firewall-based security policies, and controlled inter-VLAN communication.

Wireless network design is included within the scope, with particular emphasis on the deployment of wireless access points aggregated through the Library Distribution Switch to address distance and performance limitations associated with the Main Data Center. The scope includes secure wireless access for staff, students, dormitory users, and guests while ensuring isolation from critical hospital systems.

Network performance, reliability, and security are evaluated through simulation and testing using Cisco Packet Tracer. Centralized network monitoring and management concepts are also addressed to support proactive fault detection and performance monitoring.

The scope of this project is limited to network design, logical configuration, and simulation-based validation. Actual physical installation, real-time deployment, and operational maintenance of the network infrastructure are outside the scope of this study.

Chapter Two: Design Requirement

2.1 Design Requirement

The design requirements define the technical and functional needs that must be satisfied to build an effective campus network for Tibebe Ghion Specialized Hospital (TGSH). These requirements are derived from the hospital's operational environment, user demands, security needs, scalability expectations, and future growth considerations.

TGSH operates as both a healthcare provider and an academic institution, requiring a network that supports critical medical services, teaching, research, and administrative operations. The network must provide reliable connectivity, secure data transmission, controlled access, and high availability across all buildings and departments.

The design requirements emphasize a structured and hierarchical network architecture that supports fault isolation, simplifies management, and enables scalability. Both wired and wireless connectivity are required to support fixed devices and mobile users. The network must also support logical segmentation to separate different user groups and services while maintaining efficient communication.

Security is a key requirement due to the sensitive nature of patient and institutional data. The network design must include mechanisms for access control, traffic isolation, and protection against internal and external threats. Additionally the network should be designed to accommodate future expansion without major redesign.

2.2 Business Goal of the Project

The primary business goal of this project is to improve the reliability, security, and scalability of the Tibebe Ghion Specialized Hospital campus network in order to support efficient healthcare delivery, academic training, research activities, and administrative operations.

A well-designed network enables the hospital to provide uninterrupted medical services, protect sensitive patient data, and enhance collaboration among healthcare professionals, students, and researchers. The network also supports the hospital's long-term operational efficiency by reducing downtime, simplifying management, and allowing for future growth.

By addressing existing limitations and implementing a modern network architecture, the project aims to align the hospital's network infrastructure with its mission as a tertiary care and teaching institution.

The main business goals of the project include:

- Ensuring **continuous network connectivity** to support hospital operations.
- Improving **efficiency and coordination** among departments through reliable communication.
- Enhancing **security of institutional resources** by controlling network access.
- Supporting **academic and administrative activities** through dependable internet access.
- Reducing operational challenges caused by network failures or poor performance.
- Establishing a **scalable infrastructure** that can support future growth of the hospital.

By achieving these goals, the proposed network design contributes to improved organizational Performance and long-term Sustainability.

2.3 Technical Goals

The technical goals describe the expected performance and quality characteristics of the proposed network. These goals guide design decisions and ensure that the network meets organizational needs.

The technical goals of the network include:

- **Availability:** Maintain consistent network access for users across the campus.
- **Security:** Protect the network from unauthorized access and misuse.
- **Performance:** Provide sufficient bandwidth and acceptable response time.
- **Scalability:** Allow the network to expand without major redesign.
- **Manageability:** Enable centralized configuration and monitoring.
- **Adaptability:** Support changes in technology and user requirements.
- **Affordability:** Achieve design objectives using realistic resource utilization.

2.3.1 Technical Goals and Tradeoff

While achieving the technical goals, certain tradeoffs must be considered. Higher performance and redundancy may increase cost and complexity, while simpler designs may limit scalability and fault tolerance. The project balances these tradeoffs by selecting technologies that provide sufficient performance and security without unnecessary complexity.

Table 1: Technical Goals and Priority

Technical Goal	Priority (%)
Availability	20
Security	20
Performance	15
Scalability	15
Manageability	10
Reliability	10
Adaptability	5
Affordability	5
Usability	5
Total	100

Higher priority is given to availability and security because uninterrupted and secure connectivity is critical for hospital operations.

2.4 User communities and data stores

Understanding user communities and data stores is essential for designing a network that meets the diverse needs of TGSH. Different users have different access requirements, priorities, and security needs, while data stores require appropriate protection and availability.

2.4.1 Users of the network

The campus network of Tibebe Ghion Specialized Hospital (TGSH) serves a wide range of users with different roles, responsibilities, and access requirements. These users rely on the network for healthcare delivery, academic activities, research, and administrative operations. Identifying the different user groups is essential for designing a network that ensures security, efficiency, and appropriate access control.

The primary users of the TGSH network include clinical staff, academic staff, students, researchers, and administrative personnel. Clinical staff such as doctors, nurses, and laboratory technicians depend on the network for accessing electronic medical records, diagnostic systems, and other critical healthcare applications. These users require high availability, low latency, and strong security due to the sensitivity of patient data.

Academic staff and students use the network to support teaching, learning, and training activities. Their usage includes access to learning management systems, digital libraries, research platforms, and internet services. This user group requires reliable connectivity and controlled access to academic resources.

Researchers utilize the network for data analysis, collaboration, and access to research databases. Their activities may involve high data transfer volumes and require stable and secure network connections.

Administrative users rely on the network for day-to-day operations such as finance, human resources, scheduling, and communication. While their applications may not be as latency-sensitive as clinical systems, they still require secure and consistent network access.

In addition to internal users, the network also supports guest users, such as visitors and trainees, who require limited and controlled access, primarily to internet services. Proper identification of these user groups enables the implementation of logical segmentation and access policies that protect critical systems while meeting user connectivity needs.

2.4.1.1 Number of Users and Priority Levels

The Tibebe Ghion Specialized Hospital (TGSH) campus network supports a large and diverse population of users whose access priorities vary based on their roles and operational importance. Estimating the number of users and defining their priority levels is essential for ensuring proper bandwidth allocation, quality of service, and network reliability.

The highest priority users of the network are clinical staff, including doctors, nurses, laboratory technicians, and other healthcare professionals. These users rely on the network for real-time access to electronic medical records, diagnostic systems, laboratory results, and other mission-critical healthcare applications. Network availability, low latency, and security are critical for this user group, as any disruption may directly affect patient care.

The second priority group includes academic staff and researchers. These users depend on the network for teaching activities, research collaboration, data analysis, and access to academic resources. While their applications are important, they are generally less time-sensitive than clinical systems and can tolerate minimal delays.

The third priority group consists of students, including medical students and trainees. Their network usage primarily involves access to learning management systems, digital libraries, and internet-based educational resources. Although reliable connectivity is important, their traffic is assigned a lower priority compared to clinical and academic staff.

The fourth priority group includes administrative staff, who use the network for routine operational tasks such as finance, human resources, and general communication. These applications require stable access but are not typically latency-sensitive.

The lowest priority group is guest and visitor users, who are provided with restricted network access mainly for internet connectivity. Their traffic is isolated from internal hospital systems and assigned the lowest priority to ensure that critical services are not affected.

By categorizing users according to priority levels, the network design can effectively implement traffic control, logical segmentation, and quality of service policies that ensure critical hospital operations receive precedence over non-essential network usage.

Table 2: Number of Users and Priority Levels in access type

User Group	Priority Level	Access Type
Clinical Staff	Very High	Internal services and Internet
ICT Administrators	High	Full network access
Academic Staff	High	Internet access
Researchers	High	Internet access
Administrative Staff	Medium	Internal services
Students	Medium	Internet access
Dormitory Users	Low	Internet only
Guest Users	Very Low	Internet only

2.4.2 Data stores

Data stores represent the repositories where critical information is generated, Processed, and stored within the Tibebe Ghion Specialized Hospital (TGSH) campus network. Identifying these data stores is essential for designing a secure and reliable network infrastructure that protects sensitive information and ensures availability for authorized users.

The primary data stores within TGSH include electronic medical record (EMR) systems, laboratory and diagnostic databases, academic and research data repositories, and administrative information systems. Medical data stores contain highly sensitive patient information such as diagnoses, treatment records, laboratory results, imaging data, and clinical histories. These data stores require the highest levels of security, access control, and continuous availability.

Academic and research data stores support teaching, learning, and scientific research activities. These include digital libraries, learning management systems, research datasets, and academic content repositories. Although these data stores are important for the hospital's academic mission, they are logically separated from clinical data to maintain confidentiality and reduce security risks.

Administrative data stores manage operational information such as financial records, human resource data, inventory management systems, and internal communications. These systems require data integrity, reliability, and controlled access but are generally less time-sensitive than clinical systems.

All data stores are hosted within the main data center and mini data centers distributed across the campus. Access to these data repositories is controlled using VLAN segmentation, firewall policies, and authentication mechanisms. This approach ensures that sensitive data is protected while remaining accessible to authorized users.

Data Types

The Hospital handles text-based documents, images, research-related files, and system configuration data used for network and system management.

Data Sources

These data originate from clinical units, academic departments, and administrative offices within the hospital.

Storage Requirements

The Proposed design assumes centralized storage systems that provide controlled access, data protection mechanisms, and sufficient capacity to support organizational needs and future growth.

2.5 Network Applications

Network applications define how users interact with the campus network and determine the performance, security, and reliability requirements of the overall design. At Tibebe Ghion Specialized Hospital (TGSH), network applications support clinical services, academic activities, research operations, and administrative functions. Identifying these applications is essential to ensure that the network infrastructure can meet operational demands and prioritize critical services.

The TGSH network is designed to support both mission-critical healthcare applications and non-critical academic and administrative applications. Clinical applications require high availability, low latency, and strong security controls, while academic and administrative applications require reliable and scalable connectivity.

The network applications used within TGSH can be broadly categorized into healthcare systems, academic and research systems, administrative systems, and general network services. Each category places different requirements on bandwidth, latency, and security, which are considered during the network design process.

Healthcare-related applications include systems that support patient care, diagnostics, and clinical operations. These applications are highly sensitive and must be protected against unauthorized access while remaining continuously available to authorized medical staff.

Academic and research applications support teaching, learning, training, and scientific research. These applications require stable access to digital resources, collaboration platforms, and online services but are logically separated from clinical systems to maintain data confidentiality.

Administrative applications support daily hospital operations such as finance, human resources, inventory management, and internal communication. While these applications are important for operational efficiency, they are generally less latency-sensitive than clinical systems.

In addition, the network supports general services such as internet access and internal communication tools. These services are controlled through access policies and logical segmentation to ensure that they do not interfere with critical hospital operations.

2.5.1 Network system applications

Network system applications are the core services that enable communication, data access, and resource sharing across the Tibebe Ghion Specialized Hospital (TGSH) campus network. These applications form the foundation of daily hospital operations and support clinical, academic, research, and administrative activities.

One of the most important network system applications in TGSH is the Electronic Medical Record (EMR) system, which allows authorized medical staff to access patient records, laboratory results, imaging data, and treatment histories. This system requires high availability, strong security controls, and reliable network connectivity to ensure uninterrupted patient care.

Laboratory and diagnostic systems are also critical network applications. These systems enable the transfer of test results, imaging data, and reports between laboratories, departments, and clinicians. Because these applications handle sensitive medical data, they require secure transmission and low latency.

The network also supports academic and learning management systems, which are used by instructors and students for teaching, training, and educational activities. These systems provide access to course materials, assessments, digital libraries, and academic resources.

Research applications rely on the network for data analysis, collaboration, and access to research databases. These applications may involve moderate to high data transfer volumes and therefore require stable and efficient network connectivity.

Administrative systems, including finance, human resource management, inventory control, and internal communication platforms, are supported by the network to ensure smooth daily operations. While these systems are less time-sensitive than clinical applications, they still require secure and consistent access.

In addition, the TGSH network provides general network services such as file sharing, authentication services, and controlled internet access. These services are managed through logical segmentation and access control policies to ensure that they do not interfere with critical hospital and academic applications.

2.6 Network Application Traffic Characteristics

Understanding Network application traffic characteristics is essential for designing a reliable, secure, and scalable campus network for Tibebe Ghion Specialized Hospital (TGSH). Different applications generate different types of traffic, each with unique requirements related to bandwidth, latency, reliability, and security. Proper analysis of these characteristics ensures that critical services receive priority and that network resources are efficiently utilized.

Clinical applications, such as electronic medical record systems and laboratory information systems, generate transaction-based traffic that is highly sensitive to latency and availability. Although these applications do not always consume high bandwidth, they require consistent and reliable connectivity. Any delay or packet loss may negatively impact patient care, making these applications the highest priority in the network.

Diagnostic and imaging-related traffic involves larger data transfers, especially when handling medical images and test results. This type of traffic requires higher bandwidth and stable

connections but can tolerate slightly higher latency compared to real-time clinical transactions. Proper traffic management is necessary to prevent congestion during peak usage periods.

Academic and research applications produce moderate to high volumes of data traffic depending on activities such as content access, file downloads, and research data analysis. These applications are less latency-sensitive but require sufficient bandwidth to ensure smooth access to digital learning resources and research platforms.

Administrative application traffic typically consists of low to moderate bandwidth usage and is not highly time-sensitive. These applications include database access, record management, and internal operational systems.

While important, their traffic is assigned lower priority than clinical and academic applications. Wireless network traffic, especially from student, dormitory, and guest users, can generate unpredictable and bursty traffic patterns. This traffic primarily consists of internet browsing and general data usage. To protect critical hospital services, wireless traffic is logically segmented and assigned appropriate priority levels.

By analyzing and categorizing network application traffic characteristics, the TGSN network design can implement effective traffic prioritization, logical segmentation, and capacity planning. This approach ensures that mission-critical healthcare services remain reliable while supporting academic, administrative, and general network usage.

Table 3: Network Application Traffic Characteristics

Application Type	Traffic Model	Protocol	Bandwidth Requirement
Internal services	Client–Server	TCP/IP	Medium
Internet access	Client–Server	HTTP/HTTPS	Medium
Network management	Server–Server	SNMP, Syslog	Low
Guest access	Client–Server	HTTP/HTTPS	Low

Traffic control mechanisms ensure that essential network services are maintained during periods of high usage.

Chapter Three: Design

3.1 Design summary

This chapter presents the overall network design for Tibebe Ghion Specialized Hospital (TGSH), focusing on creating a secure, reliable, and scalable campus network that supports healthcare services, academic activities, research, and administrative operations. The design is based on the analysis of existing network limitations and the operational requirements of a large tertiary hospital and teaching institution.

The proposed network design follows a **hierarchical campus architecture**, consisting of **core, distribution, and access layers**. This structure improves scalability, simplifies network management, and enhances fault isolation. High-speed backbone connectivity is provided through the core layer, while the distribution layer aggregates traffic from different buildings and enforces routing and security policies. The access layer provides direct connectivity to end-user devices and wireless access points.

An **extended star topology** is adopted to interconnect major buildings, including hospital blocks, staff buildings, research centers, libraries, and dormitories. Fiber optic links are used for backbone connections, while copper cabling supports end devices within buildings. This topology ensures efficient traffic flow, redundancy, and ease of future expansion.

Network security is a key focus of the design. Logical segmentation using **VLANs centrally managed at the core layer** separates clinical, academic, administrative, student, dormitory, guest, and management traffic. A dedicated firewall is deployed at the network edge to provide perimeter security, access control, and Network Address Translation (NAT). Internal security is further strengthened through controlled inter-VLAN communication and access policies.

Wireless connectivity is integrated to support mobility for students and staff. Due to distance limitations from the main data center, wireless access points are physically sourced from the **library distribution switch**, enabling reliable deployment in high-demand areas such as the library and dormitories. Wireless traffic is isolated from critical hospital systems to maintain security.

The design also considers physical infrastructure components such as racks, patch panels, UPS systems, and mini data centers to ensure organized deployment, power stability, and operational reliability. Overall, the proposed network design provides a robust foundation that meets current requirements and supports future growth of Tibebe Ghion Specialized Hospital.

3.2 Design Detail

This section presents the detailed design considerations of the proposed Tibebe Ghion Specialized Hospital (TGSH) campus network. It explains how the selected topology, switching and routing mechanisms, security controls, addressing scheme, and management approach are applied to meet the project objectives. The design decisions are guided by scalability, security, reliability, and operational efficiency requirements identified in earlier chapters.

The network design adopts a modular and hierarchical structure to simplify deployment, troubleshooting, and future expansion. Each design component is discussed in dedicated subsections to clearly illustrate how the overall architecture functions as an integrated system.

3.2.1 Topology Design

The proposed network topology for Tibebe Ghion Specialized Hospital (TGSH) is designed to ensure high availability, scalability, and efficient traffic flow across the campus. An **extended star topology** combined with a **hierarchical three-layer architecture** is adopted to meet the operational and academic requirements of the hospital environment. A hierarchical topology was selected due to its scalability, ease of management, fault isolation capability, and suitability for large campus environments.

Full Campus Network Topology with Racks and ASWs

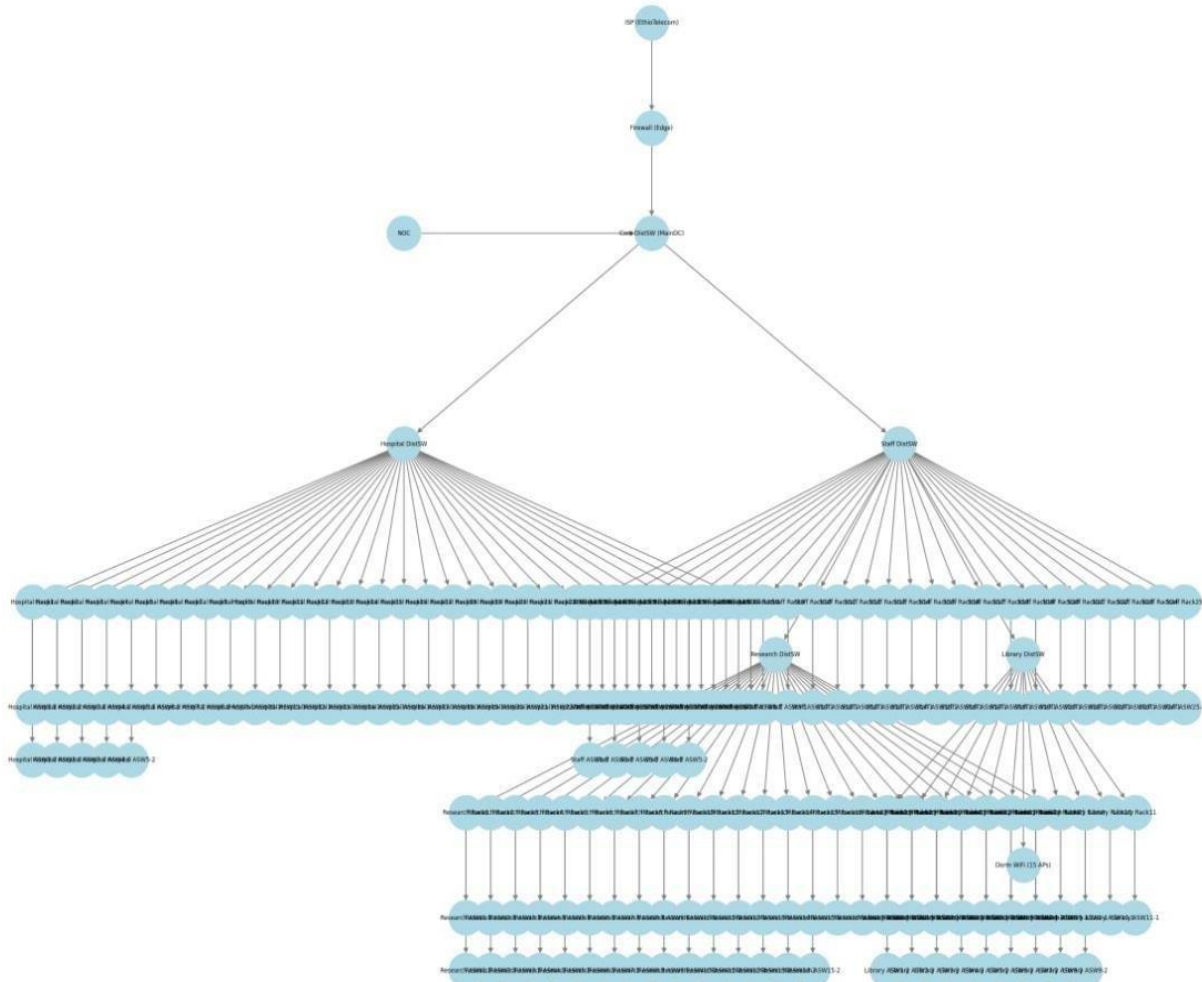


Figure 1. Full Tibebe Ghion Logical Network topology

At the core of the topology is the **main data center**, which hosts the core switches responsible for high-speed backbone connectivity and centralized control functions such as VLAN configuration and inter-VLAN routing. The core layer interconnects all major buildings using **redundant fiber-optic links** to minimize latency and provide resilience against link failures.

In addition to the main data center, **mini data centers are deployed within key buildings**, including the hospital departments, library, staff offices, and research center. These mini data centers house **distribution and access switches**, enabling localized traffic handling, improved performance, and efficient fault isolation. By distributing network resources closer to end users while maintaining centralized management from the main data center, the design enhances scalability, reliability, and overall network efficiency.

Main DC

Internet Edge & Core Flow:

The Ethio Telecom fiber cable terminates directly on the firewall at the Main Data Center. From the firewall, traffic flows to the Main-DC Core (Distribution) Switch and from the Core/Distribution switch to each distribution switch at each mini data center and flows to each access switches at the racks then to the outlets at each room. The APs are linked from the Core/Distribution switch.



Figure 2. Main data center

Mini data centers

Mini data centers are deployed within key buildings such as the hospital departments, library, staff offices, and research center to support localized network operations. Each mini data center hosts distribution and access switches that connect end-user devices within the building to the core network. This approach reduces backbone traffic, improves network performance, and allows faster fault isolation by limiting the impact of failures to a specific building. The mini data centers are centrally managed from the main data center, ensuring consistent security policies, VLAN configurations, and efficient network administration while supporting scalability and future expansion.

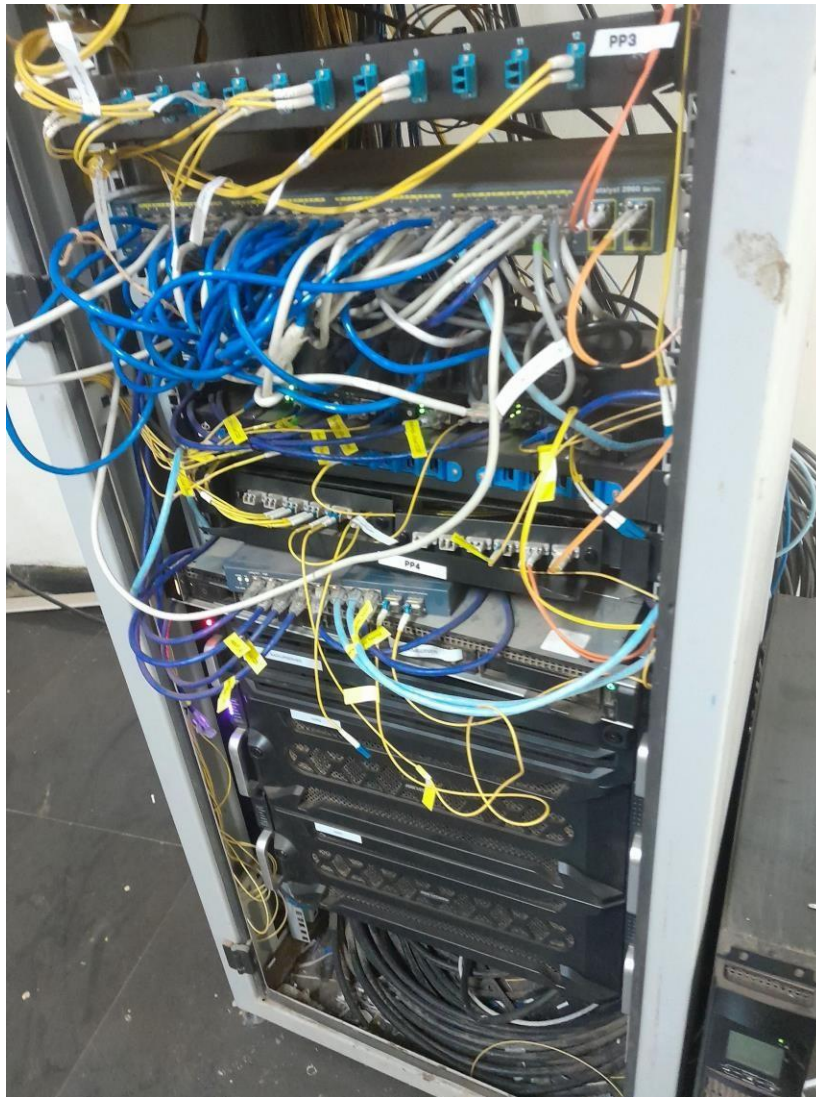


Figure 3. Mini data center

Each major building including the hospital blocks, staff buildings, research center, and library is connected to the core layer through **distribution switches** located in mini data centers at the ground floor of the respective buildings. These distribution switches aggregate traffic from access switches, enforce routing policies, and provide localized fault isolation. This design ensures that failures within one building do not affect the entire campus network.

The **access layer** consists of managed Layer 2 switches deployed in a rack-by-rack manner on each floor. These switches provide direct connectivity to end devices such as desktop computers, printers, IP-based medical equipment, and wireless access points. Power over Ethernet (PoE) is used to support wireless access points and other compatible devices, reducing additional power cabling requirements.

Wireless connectivity is integrated into the topology to support mobility for students and staff. Although wireless services originate logically from the main data center, the physical deployment of wireless access points serving dormitories and nearby facilities is sourced from the **library distribution switch** due to distance limitations from the main data center. This approach ensures reliable wireless performance while maintaining centralized control and security.

Overall, the selected topology provides a balanced design that supports efficient data transmission, simplifies network management, and allows seamless expansion as the hospital continues to grow.

3.2.2 Switching and Routing Protocol Design

The switching and routing protocol design ensures efficient data communication, reliable connectivity between network segments, and simplified network management. The selected protocols align with the hierarchical topology and support the hospital's operational and security requirements.

Switching operations are implemented at the access and distribution layers. Access layer switches operate at Layer 2 and provide connectivity to end devices. VLANs are configured to logically separate different user groups and services. Trunk links using IEEE 802.1Q tagging are deployed between switches to carry multiple VLANs over shared physical links.

Routing is required to enable communication between different VLANs and to provide access to external networks. Inter-VLAN routing is implemented using Layer 3 switching at the distribution or core layer. Routing policies are applied to regulate traffic between VLANs and ensure that only authorized communication is permitted.

Static routing is used for directing traffic between the internal hospital network and external networks. Static routing is selected because it is simple, predictable, and suitable for a controlled campus environment where network paths are stable.

Dynamic Host Configuration Protocol (DHCP) is centrally configured at the core layer to automatically assign IP addresses and network configuration parameters to connected devices. Centralized DHCP reduces manual configuration effort and minimizes addressing errors.

3.2.3 Security Design

The security design for Tibebe Ghion Specialized Hospital is developed to protect network resources, ensure controlled access, and maintain the confidentiality, integrity, and availability of information. A layered security approach is adopted, where multiple security mechanisms are applied at different layers of the network to reduce vulnerabilities and enhance protection.

3.2.3.1 Physical Security

Physical security is applied to protect critical network infrastructure. Core switches, distribution switches, access switches, servers, and firewall devices are installed in secure and restricted locations. Access to network equipment rooms is limited to authorized ICT personnel.

Critical devices are protected using uninterruptible power supply (UPS) systems to ensure continuous operation during power outages. Proper ventilation and environmental controls are implemented to protect equipment from overheating, dust, and humidity.

3.2.3.2 Security Policies

Logical security mechanisms are implemented to control access to network resources. VLAN-based segmentation is used to separate clinical staff, administrative users, students, guest users, IoT devices, voice systems, and network management traffic. This segmentation limits unnecessary communication between network segments and reduces the risk of unauthorized access.

Inter-VLAN routing policies are applied to allow only authorized traffic between VLANs. Guest and student networks are isolated from critical hospital systems to protect sensitive data and services.

Perimeter Security

Perimeter security is enforced through a dedicated firewall deployed between the internal network and the ISP. The firewall filters inbound and outbound traffic based on defined security policies, controls internet access, and protects the internal network from external threats.

Network Device and Management Security

Access to network devices is restricted to authorized administrators. Management traffic is logically separated from user traffic to prevent unauthorized access. Logging and monitoring mechanisms are implemented to detect abnormal activities and support timely incident response.

3.2.4 Addressing Design

The addressing design provides a structured, secure, and scalable IP addressing framework for the hospital network. The design separates internal private addressing from external public addressing and applies subnetting aligned with VLAN-based segmentation.

Private IP Addressing

Private IP addresses are used for all internal communications to ensure security and manageability. The 172.16.0.0/16 Class B range is selected because it offers a large number of addresses suitable for multiple VLANs across hospital buildings, supporting thousands of devices and allowing future expansion without redesigning the network. Using private IPs prevents direct access from the internet while allowing controlled external connectivity via NAT.

To optimize IP address usage, VLSM is applied to allocate subnet sizes according to the number of hosts in each VLAN. Large VLANs such as Students receive /23 subnets, while smaller VLANs like Network Management use /28 subnets. This ensures efficient utilization of IP addresses and supports future scalability.

Table 4: VLAN-Based IP Address Allocation Using VLSM

VLAN	Purpose	Users	Network	Subnet Mask	Usable IPs
10	Clinical Staff	200	172.16.10.0	/24	254
20	Academic/Admin	80	172.16.20.0	/25	126
30	Students	300	172.16.30.0	/23	510
40	Guest	40	172.16.40.0	/26	62
50	IoT/Cloud	50	172.16.50.0	/26	62
60	Network Management	10	172.16.60.0	/28	14
70	Server	15	172.16.70.0	/28	14

Public IP Addressing

Public IP addresses are assigned by the Internet Service Provider (ISP) and are used solely for external communication. In this design, the external firewall interface is configured with a public IP (e.g., 102.45.78.10/30), while all internal devices use private IPs. NAT ensures that internal devices can access external services securely, conserving the limited public IP address space and preventing direct exposure of the internal network.

3.2.4.1 Network Device Naming

A consistent and structured network device naming convention is adopted for Tibebe Ghion Specialized Hospital to simplify network management, monitoring, troubleshooting, and documentation. A standardized naming scheme helps network administrators easily identify the role, location, and function of each device within the network.

The naming convention follows a hierarchical format that reflects the organization name, device role, and device sequence number. The general format used is:

TGSH--<Location/Layer>-

Where:

TGSH represents Tibebe Ghion Specialized Hospital

indicates the type of device (Core Switch, Distribution Switch, Access Switch, Router, Firewall, Server, Wireless Access Point)

<Location/Layer> represents the network layer or physical location

is a unique numeric identifier

Examples of network device naming used in the proposed design include:

- TGSH-CORE-SW-01 – Core Layer Switch
- TGSH-DIST-SW-01 – Distribution Layer Switch
- TGSH-ACC-SW-01 – Access Layer Switch
- TGSH-RTR-01 – Edge Router

- TGSF-FW-01 – Firewall
- TGSF-SRV-01 – Application or File Server
- TGSF-WAP-01 – Wireless Access Point

This naming convention ensures consistency across the network infrastructure and supports efficient network administration, fault identification, and future expansion.

3.2.5 Network Management Design

Effective network management is essential to ensure reliable operation, security, performance, and availability of the Tibebe Ghion Specialized Hospital (TGSF) network. The proposed network management design follows the FCAPS model, which includes Fault Management, Configuration Management, Accounting Management, Performance Management, and Security Management. This structured approach enables proactive monitoring, efficient troubleshooting, and controlled administration of the hospital network.

Fault Management focuses on detecting, isolating, and resolving network faults in a timely manner. Network devices such as switches, firewalls, and servers are continuously monitored to identify failures, link disruptions, or abnormal behavior. Alerts and logs are generated to notify network administrators of faults, allowing quick response and minimizing network downtime, which is critical for hospital operations.

Configuration Management ensures that all network devices are configured correctly and consistently. Device configurations are centrally documented and backed up to prevent configuration loss and support rapid recovery in case of device failure. Any configuration changes are carefully planned and documented to avoid misconfiguration and service disruption.

Accounting Management is used to track network resource usage and user activity. Bandwidth consumption and access patterns are monitored to ensure fair resource utilization and to support capacity planning. This information also helps identify abnormal usage that may indicate misuse or security threats.

Performance Management focuses on maintaining acceptable levels of network performance. Key performance indicators such as bandwidth utilization, latency, packet loss, and device resource usage are monitored regularly. Performance analysis helps identify bottlenecks and supports optimization of network resources to ensure reliable delivery of critical healthcare and administrative services.

Security Management ensures continuous protection of network resources and data. Access to network devices is restricted to authorized personnel, and security logs are monitored to detect potential threats or unauthorized activities. Security policies are enforced consistently across the network to maintain confidentiality, integrity, and availability of hospital information systems.

3.2.5.1 Network Management Architectures

The network management architecture for TGSB consists of three main components: managed devices, agents, and a centralized Network Management System (NMS).

Managed devices include switches, firewalls, servers, wireless access points, and other network-enabled devices deployed across the hospital campus. These devices generate management information related to their status and performance.

Agents are software components running on managed devices that collect management data and communicate with the NMS using standard management protocols. These agents provide real-time information about device health, traffic statistics, and configuration status.

The Network Management System (NMS) acts as a centralized platform that collects, processes, and displays management data from all network devices. The NMS provides a unified view of the network, enabling administrators to monitor performance, detect faults, and manage configurations efficiently.

Monitoring strategies include real-time monitoring, threshold-based alerts, and periodic performance reporting. These strategies enable proactive management and help prevent service disruptions before they impact hospital operations.

NOC (Network Operations Center)

Adjacent to Main-DC centralizes control of the entire Tibebe-Ghion network, ensuring smooth operation, quick fault resolution, and security management. Monitors all devices in the network (switches, routers, firewalls, APs, servers). Detects issues like outages, high traffic, or failures in real-time. Incident Response & Troubleshooting and quickly identifies and resolves network problems. Minimizes downtime and maintains reliable services.



Figure 4: NOC Room

3.2.5.2 Network Management Tools

To support effective monitoring, control, and maintenance of the Tibebe Ghion Specialized Hospital (TGSH) campus network, standard and widely accepted network management tools and protocols are incorporated into the design.

Simple Network Management Protocol (SNMP) is used to collect performance statistics, interface status, and device health information from network devices across all layers. SNMP enables centralized visibility and proactive issue detection.

Secure Shell (SSH) is employed for secure remote administration of network devices. Encrypted communication ensures that configuration and management access is protected from interception and unauthorized use. Syslog is utilized for centralized logging of system events, security alerts, and operational messages generated by network devices. Centralized logging supports troubleshooting, auditing, and security analysis.

Together, these tools provide comprehensive monitoring, secure administration, and effective fault management, ensuring stable operation and long-term reliability of the TGSH campus network.

Chapter four: Network Implementation

4.1 Overview of Network Implementation

This chapter presents the implementation of the proposed campus network design for Tibebe Ghion Specialized Hospital (TGSH). The implementation phase translates the logical and physical designs described in previous chapters into a functional network infrastructure that supports clinical, academic, research, and administrative operations.

The network is implemented using a hierarchical architecture consisting of core, distribution, and access layers. This layered approach ensures scalability, fault isolation, efficient traffic management, and ease of administration. Core network devices provide centralized routing, VLAN configuration, and policy enforcement, while distribution and access layer devices support building-level and end-user connectivity.

Both wired and wireless network components are deployed to ensure comprehensive coverage across hospital buildings, staff facilities, research centers, library areas, and dormitories. Emphasis is placed on reliability, security, and performance to meet the operational demands of a tertiary healthcare and teaching institution.

The implementation follows industry best practices and aligns with the objectives of the final year internship project, ensuring that the network infrastructure is robust, secure, and adaptable to future expansion.

4.2 Physical Network Design

This section describes the physical implementation of the TGSH campus network, focusing on device placement, cabling infrastructure, and data center arrangements. The physical design ensures structured deployment, efficient maintenance, and high availability of network services.

A main data center serves as the central hub of the network, housing core switching, firewall, and centralized services. In addition, mini data centers are deployed at the ground floors of major buildings to host distribution switches and support localized traffic aggregation.

Structured cabling is used throughout the campus, with fiber optic links connecting the main data center to building-level mini data centers, and Category 6 Ethernet cabling used for horizontal connections to access layer devices. Network racks are securely installed and labeled to support organized equipment placement and ease of troubleshooting.

Physical Network Topology

The physical topology of the proposed TGSB campus network follows an **extended star architecture**, where all campus buildings are interconnected through a centralized core layer located at the main data center. Access switches deployed within individual buildings such as the hospital, staff offices, research center, library, and student dormitories are physically connected to distribution and core layer switches using structured cabling.

Unshielded Twisted Pair (UTP) Ethernet cables, specifically Category 6 (Cat6), are used for intra-building connections between end devices and access switches due to their reliability, cost-effectiveness, and support for high-speed data transmission. Backbone connections between distribution switches and the core switch are implemented using high-capacity links to ensure efficient and stable communication across the campus network. A dedicated firewall device provides secure connectivity between the internal network and the Internet service provider.

This physical topology accurately represents the actual placement of network devices and cabling infrastructure and provides a robust foundation for implementing logical network functions such as VLAN segmentation, inter-VLAN routing on the core switch, and centralized security enforcement.

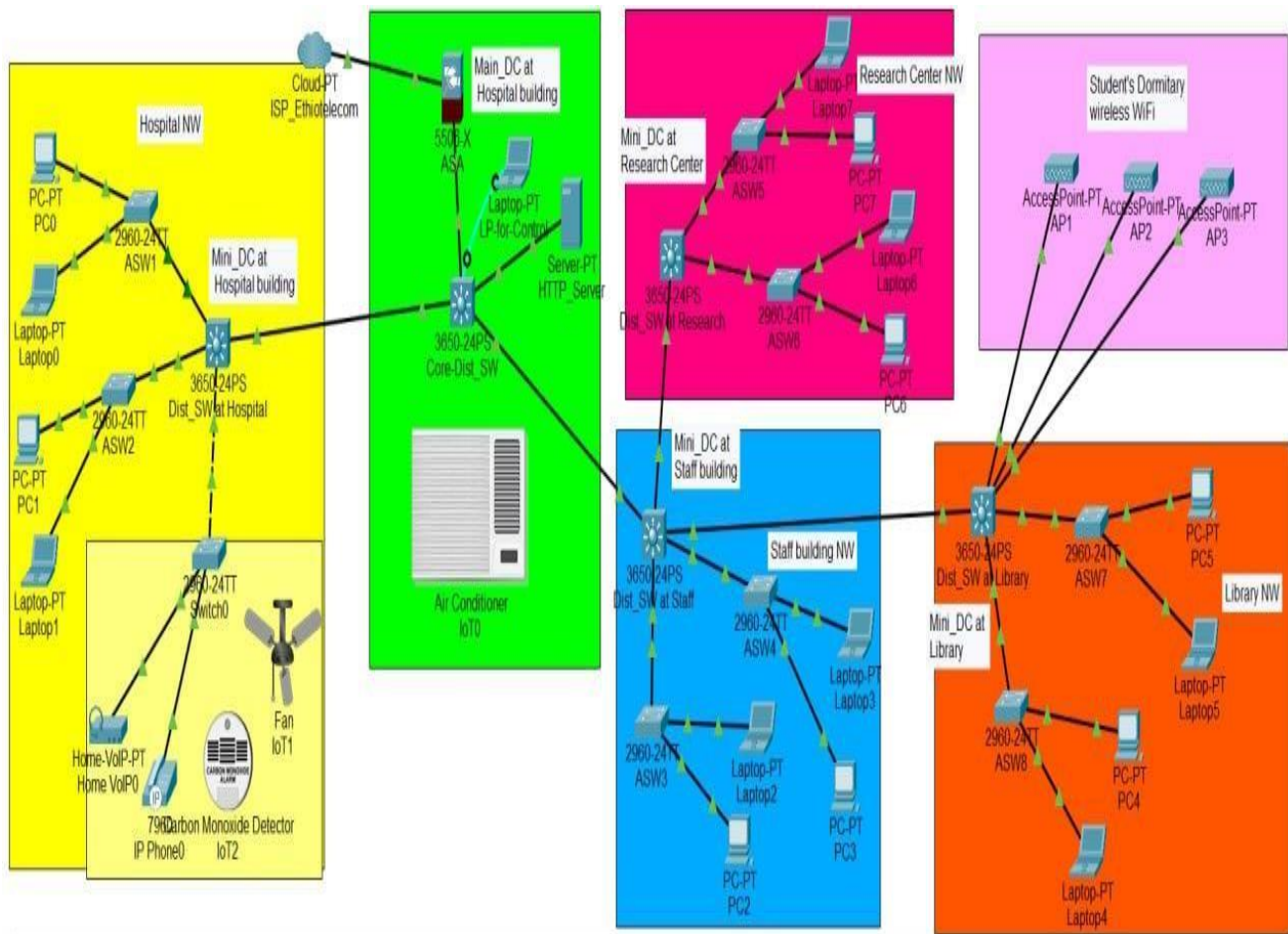


Figure 5: Physical Network Topology of TGSH Campus Network

4.2.1 Network Materials and Equipment Used

The implementation of the TGSJ campus network utilizes enterprise-grade networking equipment selected based on reliability, scalability, and compatibility with the proposed design. Core and distribution layer switching is performed using Layer 3-capable switches, while access layer connectivity is provided through managed access switches.

Table 5: Hardware and Cabling Specification for TGSJ Network

Device Category	Specification	Purpose
Core Switch	Layer 3, 24/48 ports, Gigabit/10Gb, Redundant Power	Backbone connectivity and inter-VLAN routing
Distribution Switch	Layer 3, VLAN and ACL support	Traffic aggregation and policy enforcement
Access Switch	Layer 2, PoE-enabled, 24/48 ports	End-user and device connectivity
Firewall	Next-Generation Firewall (NGFW)	Perimeter security and traffic filtering
Wireless Access Point	Dual-band (2.4/5 GHz), Enterprise-grade	Wireless connectivity and mobility
Servers	Application, File, Authentication, NMS	Hospital systems and network services
Cabling (LAN)	Cat6 UTP	High-speed internal data transmission
Cabling (Backbone)	Fiber Optic	High-speed inter-building connectivity

UPS	Online UPS	Power backup for critical devices
Racks and Patch Panels	Standard 42U racks	Equipment organization and protection

Wireless connectivity is supported through centralized wireless access points powered using Power over Ethernet (PoE). A perimeter firewall is deployed to enforce security policies and protect the internal network from external threats.

Additional materials include network racks, patch panels, fiber optic cables, Category 6 Ethernet cables, uninterruptible power supply (UPS) units, and labeling components. These materials ensure physical protection, power continuity, and structured organization of the network infrastructure.



Figure 6: Fiber optic cable

4.2.2 Rack-by-Rack Distribution Design

The rack-by-rack distribution design ensures systematic deployment of access switches across different buildings and floors within the TGSH campus. Each rack accommodates access switches, patch panels, and power units to support local user connectivity.

In the hospital buildings, racks are distributed across multiple floors in right, left, and central wings to ensure balanced coverage and reduced cable lengths. Staff buildings, research centers, and library facilities follow similar structured rack deployment strategies based on building layout and user density.

Mini data center racks located at ground floors host distribution switches that aggregate access layer traffic and forward it to the core layer through fiber optic links. This arrangement improves performance and simplifies network expansion.

Hospital:

1 L3 Dist SW; ~30 racks, 35 ASWs (1–3 per rack). Building is U shaped with Right/Left/Central wings.

Rack distribution

- ✓ G: ~8 racks (Right 3, Left 3, Center 2)
- ✓ G+1: ~8 racks (Right 3, Left 3, Center 2)
- ✓ G+2: ~7 racks (Right 3, Left 2, Center 2)
- ✓ G+3: ~7 racks (Right 3, Left 2, Center 2)
- ✓ plus 1 extra rack at Ground for the mini DC Dist SW

Staff Building

Rack distribution

- ✓ Meseret (Ground): 3 racks
- ✓ Skill Lab: 4 racks (G, G+1, G+2, G+3)
- ✓ Classrooms (U shaped): 8 racks → G:2, G+1:2, G+2:2,
- ✓ plus other classroom at G+2:2 Offices (U shaped): 10 racks → G:2, G+1:2, G+2:2, G+3:2, G+4:2
- ✓ plus 1 extra rack at Ground for the mini DC Dist SW

Research Center

Rack distribution

- ✓ 1 L3 Dist SW; ~15 racks, 40 ASWs (1–3 per rack).
- ✓ Building shaped like a standing person with left/right arms (wings); 3 parts: Center, Left Wing, Right Wing;
- ✓ floors G to G+3.
- ✓ plus 1 extra rack at Ground for the mini DC Dist SW
- ✓ Right Wing: 5 racks → G:2, G+1:1, G+2:1, G+3:1
- ✓ Left Wing: 5 racks → G:2, G+1:1, G+2:1, G+3:1
- ✓ Center: 5 racks → G:2, G+1:1, G+2:1, G+3:1

Library

Rack distribution

- ✓ 1 L3 Dist SW; 14 racks, 20 ASWs (1–3 per rack).
- ✓ 3 racks G+1:
- ✓ 4 racks; G+2:
- ✓ 4 racks; G+3:
- ✓ 2 rack; G+4:
- ✓ 1 rack plus 1 extra rack at Ground for the mini DC Dist SW

Dormitory Wi Fi

15 APs fed via PoE from nearby ASWs; fiber from Main DC (via Library) to AP aggregation.

Bill of Quantity (BoQ)

The Bill of Quantity (BoQ) provides an itemized list of the network devices, cabling, and supporting infrastructure required to implement the proposed TGSB campus network. The quantities listed are **design estimates intended to support the network topology and performance requirements** and do not represent actual purchased or deployed equipment.

These estimates ensure that sufficient resources are considered for VLAN-based segmentation, backbone connectivity, and wireless coverage across all buildings while maintaining scalability for future expansion. Prices reflect current Ethiopian market rates in 2025/2026 and include enterprise-grade equipment such as core and distribution switches, access switches, racks, patch panels, fiber optic cables, wireless access points, a firewall, and a central server. The BoQ serves as a guide for planning and cost estimation and provides a clear understanding of the infrastructure required to support the hospital and academic network environment.

Item No	Description	Unit	Quantity	Unit Price (ETB)	Total Price (ETB)	Notes / Allocation
1	Core Switch (Layer 3, 48 Ports, 10G Uplink)	Unit	1	250,000	250,000	Main data center core switch for inter-VLAN routing
2	Distribution Switch (Layer 2/3, 24 Ports)	Unit	4	120,000	480,000	One per mini data center (hospital, library, staff, research)
3	Access Switch (24-port managed)	Unit	125	43,500	5,437,500	Distributed across buildings (hospital 35, staff 25, library 15, research 20,

						dormitories 30)
4	Rack Cabinet (42U)	Unit	100	28,000	2,800,000	Racks for servers, switches, and patch panels
5	Fiber Optic Cable (Single Mode, 500 m reel)	Roll	2000	20,000	40,000,000	Backbone links connecting main and mini data centers and all buildings
6	Patch Panel (24-port)	Unit	105	12,000	1,260,000	Installed in each rack for structured cabling
7	Wireless Access Point (Indoor, PoE)	Unit	15	8,000	120,000	Wireless coverage for staff, students, and visitors
8	Rack-Mount Server	Unit	1	60,000	60,000	Centralized services in main data center
9	Firewall Appliance (Enterprise)	Unit	1	150,000	150,000	Network edge firewall for secure internet access

4.2.3 Wireless Network Implementation

Wireless network implementation provides mobility and flexible access for staff, students, and guests across the TGSB campus. Wireless access points are deployed in staff areas, library spaces, dormitories, and public zones to ensure adequate coverage and capacity.

Initially, wireless connectivity was sourced from the main data center; however, due to the physical distance between the main data center and dormitory areas, wireless distribution was shifted to the library distribution switch. This change reduces latency, improves signal reliability, and enhances overall wireless performance.

Wireless networks are logically segmented to separate internal users from guest and dormitory users, ensuring security and controlled access to hospital systems.

4.3 Logical Network Implementation

This section describes the logical configuration of the TGSB campus network. Logical implementation focuses on VLAN creation, inter-VLAN routing, traffic segmentation, and IP addressing to support security, performance, and scalability requirements.

VLANs are configured centrally at the core switch to logically separate hospital departments, staff networks, research units, library services, dormitories, guest access, server, management traffic, and specialized service networks. Inter-VLAN routing is performed at the core layer to enable controlled communication between authorized network segments.

4.3.1 IP Addressing and Subnetting Scheme

The TGSB campus network uses a **private Class B IP addressing scheme** based on the address block **172.16.0.0/16**, in compliance with RFC 1918. This large address space provides sufficient IPs to support current devices, VLAN-based segmentation, and scalable expansion for future departments and services.

To optimize IP address usage and reduce wastage, **Variable Length Subnet Masking (VLSM)** is applied. Each VLAN is assigned a subnet size appropriate to the number of devices it serves. Large VLANs such as the Students VLAN receive larger subnets (/23) to accommodate all concurrent users, while smaller VLANs such as Management or Server use smaller subnets (/28) to efficiently utilize the address space. Default gateways for all VLANs are configured at the **core switch**, where inter-VLAN routing is implemented.

End-user devices receive IP addresses dynamically through DHCP, while critical infrastructure devices, including the **core and distribution switches, firewall, servers, and wireless controllers**, are assigned static IP addresses for stable management and monitoring.

4.3.2 VLAN-to-Subnet Mapping

VLAN	Purpose	Users	Network	Subnet Mask	Usable IPs
10	Clinical Staff	200	172.16.10.0	/24	254
20	Academic/Admin	80	172.16.20.0	/25	126
30	Students	300	172.16.30.0	/23	510
40	Guest	40	172.16.40.0	/26	62
50	IoT/Cloud	50	172.16.50.0	/26	62
60	Network Management	10	172.16.60.0	/28	14

70	Server	15	172.16.70.0	/28	14
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The VLAN-to-subnet mapping table provides a clear association between logical VLANs and their corresponding IP subnets. This mapping simplifies network administration, supports consistent policy enforcement, and aids troubleshooting by clearly identifying traffic boundaries within the TGSN campus network.

4.3.3 DHCP Implementation Details

Dynamic Host Configuration Protocol (DHCP) is implemented to automate IP address assignment for end-user devices across the TGSN campus network. DHCP services reduce administrative overhead, minimize configuration errors, and ensure efficient utilization of IP address space.

DHCP pools are created for each user-facing VLAN, with address scopes aligned to the defined subnet ranges. Each pool assigns IP addresses, subnet masks, default gateways, and Domain Name System (DNS) information dynamically to client devices.

The core switch functions as the primary DHCP server for the campus network, ensuring centralized control and consistent address allocation. For VLANs where DHCP services are relayed, DHCP relay functionality is configured at the core layer to forward client requests appropriately.

Critical network infrastructure devices, including switches, firewalls, servers, and wireless access points, are excluded from DHCP pools and configured with static IP addresses to ensure stable management and monitoring.

This DHCP implementation enhances network reliability, simplifies device onboarding, and supports seamless mobility for wired and wireless users across the TGSN campus.

4.3.4 Server VLAN and Web Server Implementation

A dedicated **Server VLAN** is implemented to host centralized services required by Tibebe Ghion Specialized Hospital. This VLAN is logically separated from user and service networks to enhance security, performance, and access control.

The hospital web server is deployed as a standalone server device physically connected to the core switch within the main data center. The server is assigned a static IP address from the Server VLAN subnet (**172.16.9.0/24**) to ensure consistent accessibility and reliable service delivery.

Access to the web server is controlled through inter-VLAN routing policies and firewall rules, allowing only authorized internal networks to reach the server while preventing unauthorized access from guest and non-critical networks. This approach protects hospital data and ensures the availability of web-based clinical and administrative applications.

Placing the web server at the core layer ensures low-latency access, centralized security enforcement, and simplified management, aligning with best practices for enterprise and healthcare network deployments.

Logical policies are applied consistently across the network to ensure efficient traffic flow, reduced broadcast domains, and improved fault isolation.

4.4 Security Implementation

Security implementation is a critical component of the TGSH campus network. A layered security approach is adopted to protect sensitive clinical and academic data while ensuring service availability.

VLAN-based segmentation is enforced at the core layer to isolate critical hospital systems from non-critical user networks. A perimeter firewall is deployed to control inbound and outbound traffic, implement network address translation, and enforce access control policies.

Administrative access to network devices is restricted and secured to prevent unauthorized configuration changes. These measures collectively enhance the security posture of the TGSH network.

4.5 Network Management and Monitoring

This section describes the implementation of network management and monitoring mechanisms for the Tibebe Ghion Specialized Hospital (TGSH) campus network. The objective of this implementation is to ensure continuous network availability, efficient performance monitoring, rapid fault detection, and secure administrative control over all network components.

Network management is implemented using a centralized approach aligned with the hierarchical network architecture. A dedicated management network is logically separated from user and application traffic to protect management operations from congestion and unauthorized access. Core network devices act as aggregation points for management information collected from distribution and access layer devices.

Simple Network Management Protocol (SNMP) is implemented on all critical network devices to enable real-time monitoring of device status, interface utilization, error rates, and hardware health. SNMP monitoring supports proactive maintenance by allowing network administrators to identify potential issues before they affect hospital services.

Secure Shell (SSH) is configured for remote administration of network devices. Access is restricted to authorized administrators and limited to the management network, ensuring secure configuration and maintenance activities.

Syslog services are implemented to centralize system logs and security-related events generated by network devices. Centralized logging supports auditing, troubleshooting, and detection of abnormal behavior within the network infrastructure.

Together, these network management and monitoring mechanisms enhance operational visibility, improve response time to network incidents, and ensure stable and secure operation of the TGSH campus network in support of clinical, academic, research, and administrative functions

Stakeholder Communication Plan

Effective and structured communication with stakeholders is essential for the successful implementation of the TGSH campus network. Key stakeholders include **hospital management, ICT staff, clinical staff, administrative users, and external service providers.**

Communication activities are conducted at every stage of the project:

- **Requirement Validation:** Meetings with hospital management and departmental representatives to confirm network requirements and user needs.
- **Design Review:** Technical review sessions with ICT staff and service providers to ensure the proposed design meets operational requirements and best practices.
- **Deployment Updates:** Regular progress reports are shared with stakeholders during equipment installation, cabling, and configuration.
- **Testing & Evaluation:** Feedback sessions with users are conducted to ensure that the network meets service expectations.
- **Handover & Acceptance:** Final review and formal handover of the network infrastructure to the hospital ICT team.

Communication channels include **weekly meetings, progress emails, and formal reports**, while **feedback mechanisms** are established to address stakeholder concerns promptly. This structured communication ensures alignment with hospital objectives and smooth adoption of the new network system.

Training Plan

Training is essential to ensure that both **end users and ICT staff** can effectively utilize, maintain, and manage the network. Training programs are tailored based on user roles:

- **End Users:**

Basic sessions on network usage policies, secure access procedures, and services available on the network.

Practical demonstrations on connecting to VLANs, accessing resources, and understanding network restrictions.

Security awareness training to educate staff on phishing, unauthorized access, and proper handling of sensitive information.

- **ICT Staff:**

Advanced technical training on **network configuration**, including VLAN setup, DHCP scope management, inter-VLAN routing, firewall configuration, and VLSM IP planning.

Troubleshooting procedures for switches, routers, wireless APs, and server connectivity.

Monitoring tools and techniques to track network performance, detect faults, and respond to security incidents.

- **Ongoing Training:**

Refresher courses and workshops to update staff on network upgrades, new devices, and security practices.

Effectiveness Measurement Plan

The effectiveness of the network implementation is measured using both **technical performance indicators** and **user satisfaction metrics**:

- **Technical Performance Metrics:**

Network availability: Percentage uptime of network services.

Bandwidth utilization: Monitoring peak and average usage to detect congestion.

Latency and packet loss: Ensuring low delay and minimal loss for critical hospital applications.

Security incidents: Frequency of unauthorized access, malware, or firewall breaches.

- **User Feedback:**

Surveys and interviews to assess service quality, accessibility, and satisfaction among clinical staff, administrative users, and students.

Reports on ease of network use, wireless connectivity, and application performance.

- **Evaluation Process:**

Measurements are collected periodically during the first months after deployment.

Collected data is analyzed to determine if the network meets the design objectives and hospital service requirements.

Results guide **continuous improvement** of network policies, configuration, and user support.

List of Known Risks and Mitigation Measures

Potential risks during network deployment and operation are identified and addressed as follows:

Risk	Mitigation Measure
Hardware delivery delays	Early procurement planning, working with multiple suppliers, maintaining spare devices
Power interruptions	Deployment of UPS systems, backup generators, and redundant power lines
Budget limitations	Prioritizing critical devices, phased procurement, and cost tracking
Configuration errors	Detailed network documentation, peer review of configurations, simulation in Packet Tracer before deployment
Security threats	Firewalls, ACLs, VLAN isolation, regular security audits, user training

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Fallback Plan

A fallback plan ensures **rapid recovery and service continuity** in case of network failure or unexpected incidents:

- **Configuration Backups:** All network devices have **daily or weekly backup configurations** to allow immediate restoration.
- **Redundant Links and Devices:** Critical backbone links and core/distribution switches have redundancy to maintain connectivity during failures.
- **Alternative Access:** If external internet connectivity fails, temporary solutions such as secondary ISP links or mobile hotspots are available for critical services.
- **Stepwise Recovery:** In the event of a failure, the ICT team follows a **documented recovery procedure**, restoring devices and connections in a prioritized order to minimize operational impact.
- **Monitoring:** Continuous monitoring tools detect issues early, allowing proactive mitigation before full service disruption occurs.

Plan for Evolving the Design

The network design includes provisions for future growth and technological advancements. Modular expansion is supported by the hierarchical architecture, allowing new switches, VLANs, and services to be added without major redesign. The design supports future integration of new medical systems, increased IoT deployment, higher bandwidth requirements, and potential adoption of emerging technologies such as IPv6 and cloud-based services. Project schedule

The project schedule outlines the major activities carried out during the design and implementation of the secure Tibebe Ghion Campus Network Infrastructure. The schedule was developed to ensure systematic progress, effective time management, and successful completion of the project within the allocated academic period.

The project was implemented in a phased approach, beginning with problem identification and requirement analysis. During this initial phase, data was collected through observation, interviews with ICT staff, and review of existing documentation to understand the current network challenges, user requirements, and organizational needs. This phase laid the foundation for defining clear project objectives and scope.

The second phase focused on literature review and conceptual design. Relevant textbooks, research papers, and networking standards were reviewed to identify best practices in campus network design, security, and scalability. Based on this review, a hierarchical network architecture was proposed, incorporating core, distribution, and access layers.

The third phase involved detailed network design, including topology selection, VLAN segmentation, IP addressing, routing and switching design, security policies, and network management mechanisms. Logical and physical network diagrams were developed using simulation tools to visualize the proposed design.

The fourth phase covered network simulation and implementation using Cisco Packet Tracer. Network devices were configured, VLANs and routing were implemented, and security policies were tested to evaluate performance, connectivity, and fault isolation. Necessary adjustments were made based on test results.

The final phase focused on documentation and report writing. All project activities, design decisions, configurations, and results were compiled into a structured academic report. The project schedule ensured that each phase was completed sequentially and efficiently, resulting in a well-documented and technically sound final year project.

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Appendices

The project was developed in response to the challenges observed in the existing network infrastructure of Tibebe Ghion Specialized Hospital (TGSH). The current network suffers from problems such as limited reliability, lack of redundancy, weak internal security, outdated networking devices, and poor support for future expansion.

These issues affect daily hospital operations, academic activities, research work, and administrative services. To address these problems, the project proposes a modern campus-wide network design based on a structured core–distribution–access architecture.

In this design, the core layer handles high-speed routing and backbone traffic, the distribution layer manages policy enforcement and traffic control, and the access layer connects end user devices and wireless access points. This structured approach makes the network easier to manage, more reliable, and ready for growth.

This proposed system uses Virtual Local Area Networks (VLANs) to logically separate different user groups such as hospital clinical systems, staff and administration, library and research units, dormitory users, guest users, and network management.

Communication between these networks is carefully controlled using firewall rules and access policies to ensure that sensitive hospital systems remain protected. Security is further strengthened through the use of a centralized firewall that provides perimeter protection, traffic filtering, and secure internet access using Network Address Translation (NAT).

Wireless networks are designed with separate SSIDs for staff, dormitory users, and guests, ensuring safe and isolated wireless access across the campus. To improve service availability, the network design reduces single points of failure by introducing redundancy in critical network links and devices.

Centralized monitoring and management are supported through a Network Operations Center (NOC), allowing ICT staff to monitor performance, detect faults early, and manage the network more efficiently.

The proposed network design is tested and validated using simulation tools such as Cisco Packet Tracer. The design also prepares TGSN for future technologies, including IoT-based medical devices, cloud services, and VoIP systems.